

MODULE 02

RDBMS Concepts and Architecture

DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

RDBMS Concepts and Architecture

What is RDBMS?

Relational Database Management System stores data in tables with defined relationships.
Enforces ACID properties and data integrity through constraints.

ACID Properties

Atomicity: all operations succeed or all fail
Consistency: database stays in a valid state
Isolation: concurrent transactions do not interfere
Durability: committed data survives crashes

Popular Systems

MySQL: open-source web databases
PostgreSQL: advanced open-source
Oracle: enterprise grade
SQL Server: Microsoft enterprise
SQLite: lightweight embedded

3-Tier Architecture

External Level: user views
Conceptual Level: schema definitions
Internal Level: physical disk storage

Table Example

```
CREATE TABLE students (  
  id INT PRIMARY KEY,  
  name VARCHAR(100) NOT NULL,  
  dept_id INT REFERENCES departments(id)  
);
```